605.486 – Android Development

Module 13a-e

Notifications

Communicating with the User

- We've seen Dialog
  - Immediate/temporary
  - Interactive
- Toast
  - Immediate and temporary message to user
  - Non interactive
- ***Never*** *have thread/service bring up dialog/toast!*
  - Why not?
- *So how do we communicate from background tasks?*

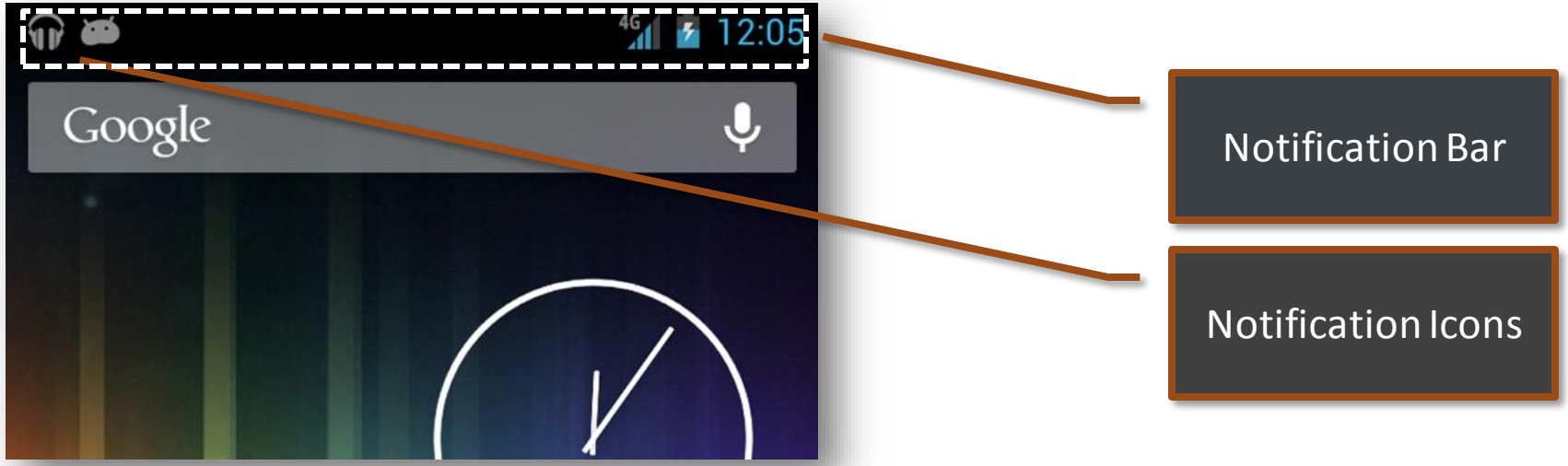
Notifications

- Appear in status at top and notification pulldown
- May contain actions user can perform
- Use PendingIntents to store a context in which to invoke actions
- Normally *not* invoked from Activities
  - Why not?
  - When might you want to?

Notification Best Practices

- Don’t be noisy!
- Collapse multiple notifications into one where possible
  - When might it be useful to actually have separate notifications?

Notification Bar



- Notification bar present unless "fullscreen"
- Icons indicate available notifications
  - Some are persistent (always remain)
  - Others disappear after viewed

Notification Bar

- ↓ Single-swipe down to see notifications
- ⇓ Double-swipe down to expand controls
  - (on newer versions of Android)

Opened Notification bar

Send Notification

- Create Intent to start desired activity when notification touched
- Create PendingIntent to bundle current context/permissions
- Create Notification with NotificationCompat
  - Icons – if no large set, small is scaled up – single icon
  - Ticker – text at top of screen when notification appears
  - Number – info for user, like number of emails
  - Content title/text – text in main section
  - Pass in Pending Intent

- Call notification service's notify() with id
  - Same id = update it
  - Different id = separate notifications

Cancel Notification

- Ask notification service to cancel(id)
  - (or call cancelAll() to remove all notifications from this application)

Example: Simple Notification

- Using buttons to display notification
  - DON’T DO THIS!!! FOR EXAMPLE ONLY!!!
- Initially notification remains when clicked
  - App can cancel, user can clear all, user can sweep
- Add .setAutoCancel(true)
- Cancel with id, or cancelAll
- Pass in unique ids
- .setOngoing(true)

Example: Expandable Notification

- Only available in JellyBean and later
- Can include image, block text or inline text
- Example
  - Setup inbox style
  - Set style in the notification
    - Expanding styles ignored for older versions
  - Need to use actual device to test expand/contract

Back Stack Issues

- When jumping into application via notification
  - May need navigation context
  - e.g: master list activity and detail activity

- Test app – create notification, exit app, click notification, back
  - Bad!

>= API 16 (JellyBean) Define "`parentActivityName`"

```
<activity
    android:name=".DetailActivity"
    android:parentActivityName=".MainActivity"
    android:label="@string/app_name" >
    <meta-data android:name="android.support.PARENT_ACTIVITY"
               android:value=".MainActivity" />
</activity>
```

- Note meta-data – used by compat library

Back Stack Issues (2)

- Build back stack

```
TaskStackBuilder stackBuilder = TaskStackBuilder.create(this);

stackBuilder.addParentStack(DetailActivity.class);
stackBuilder.addNextIntent(detailIntent);

PendingIntent detailPendingIntent =
                    stackBuilder.getPendingIntent(
                          0, PendingIntent.FLAG_UPDATE_CURRENT );
```

- Test – now see that the MainActivity appears on back!

Actions in Notifications

- Only available in JellyBean and later
- ACTIONS MUST BE ACCESSIBLE THROUGH OTHER MEANS
  - Older versions can't run them otherwise
- Up to three actions can be added to notification
  - .addAction(icon, title, intent)